

Manipulation of Articulated Objects Using Trajectory Primitives Learned Through Simulation

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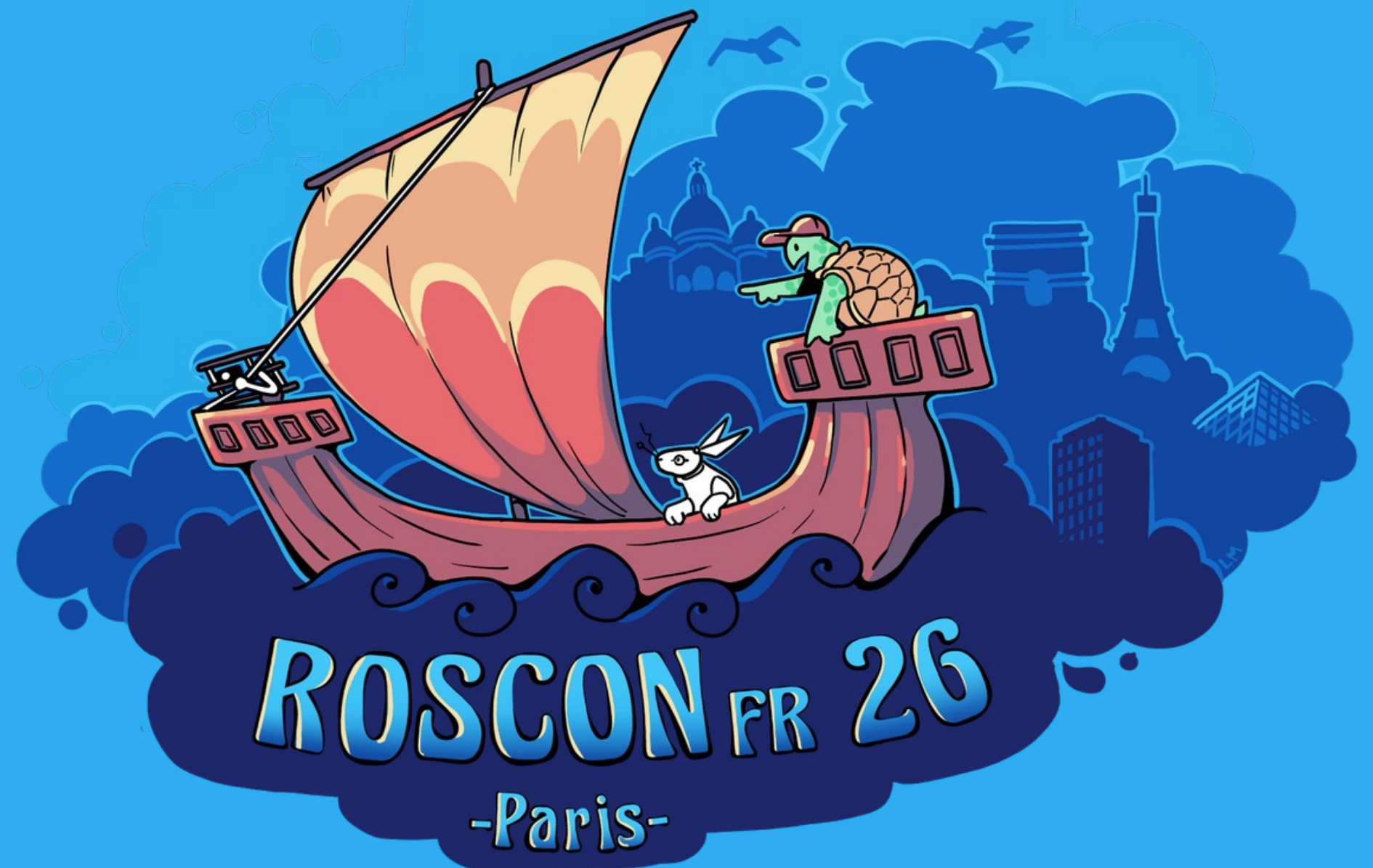
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●●●● ROSCon FR
●●●● Paris



24/06/26, Paris

Introduction to Robotic Manipulation in Real World Environments

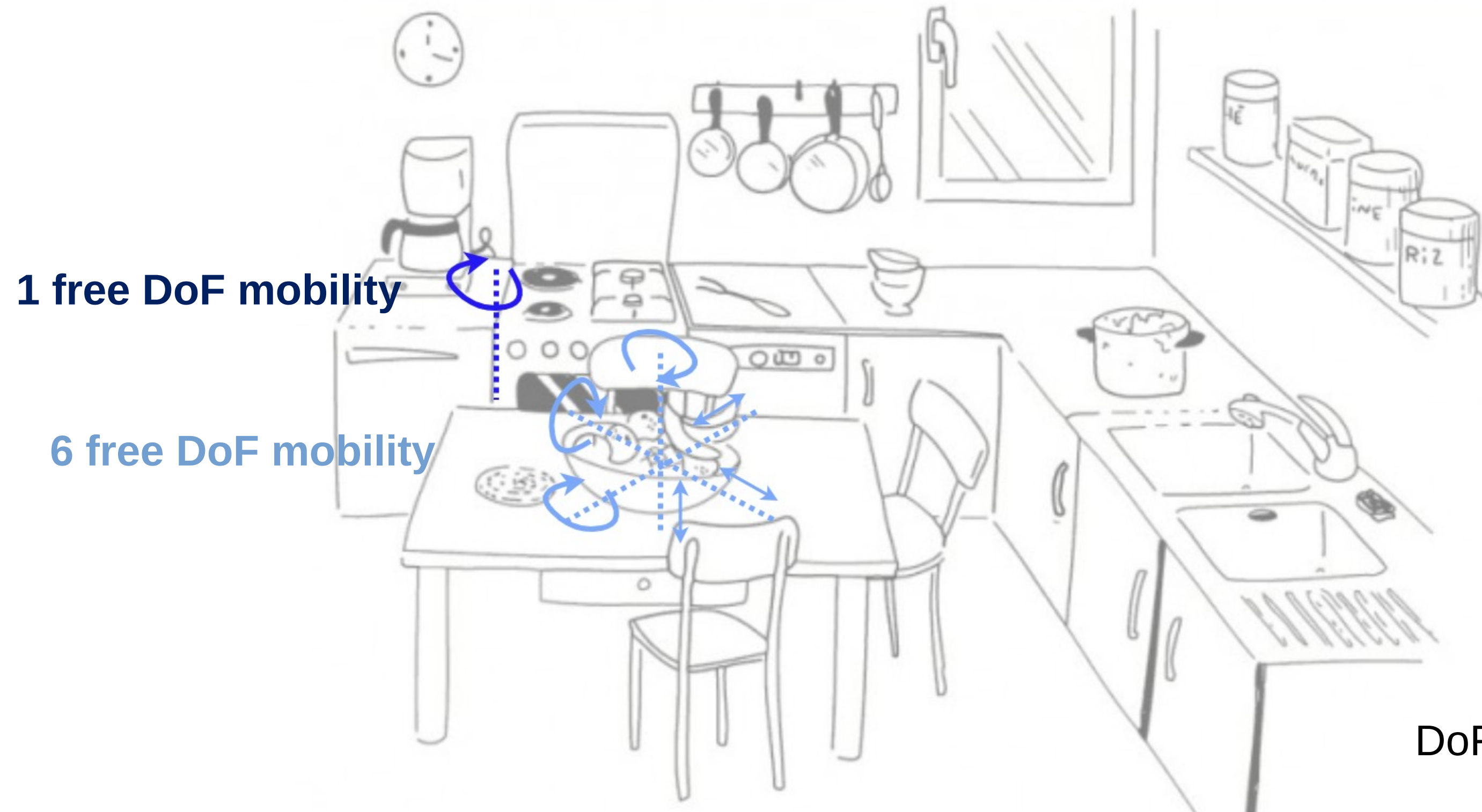
- Manipulation → **Physical contact interaction** between end-effector & environment
- Open-ended environment → X scenarios, X physical configurations ...



How can a robot learn to manipulate any real environment?

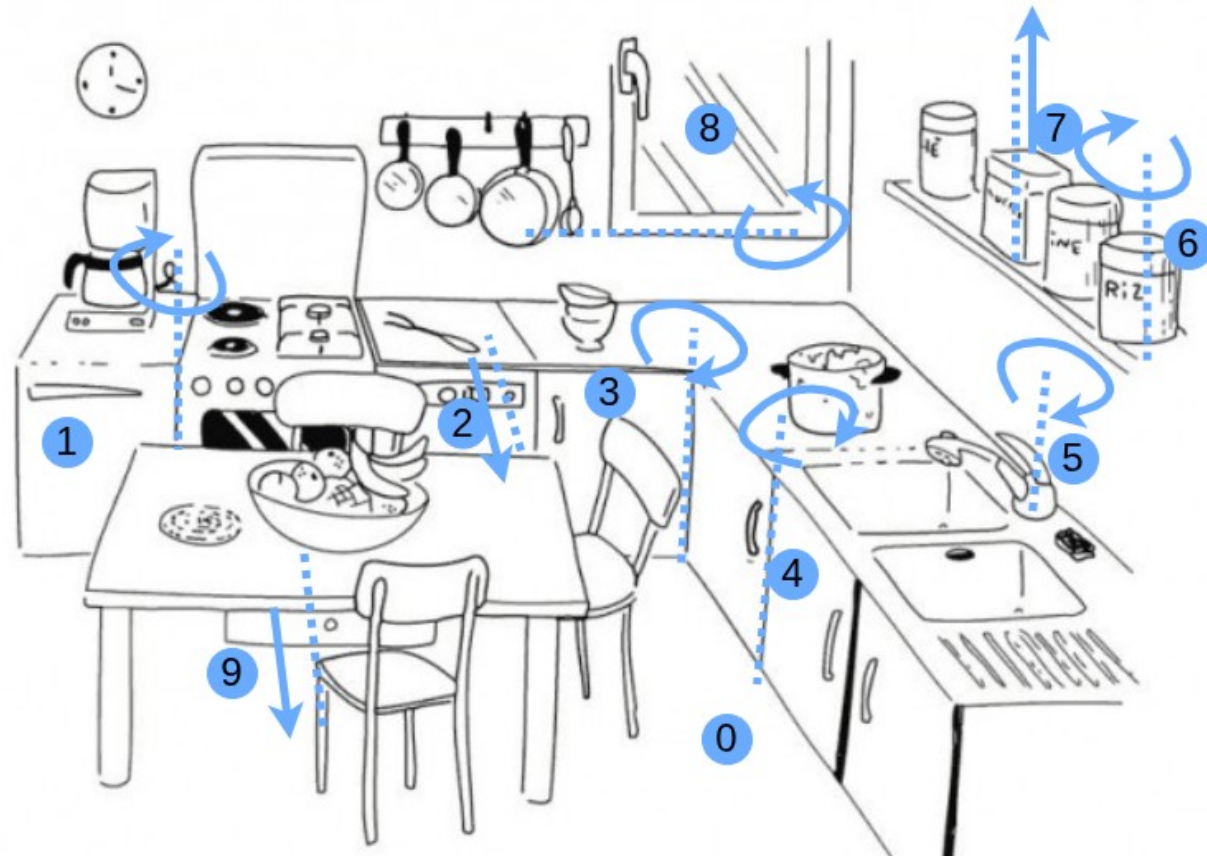
Main learning Challenge : Sparse reward environment

Robotic manipulation on constrained environment

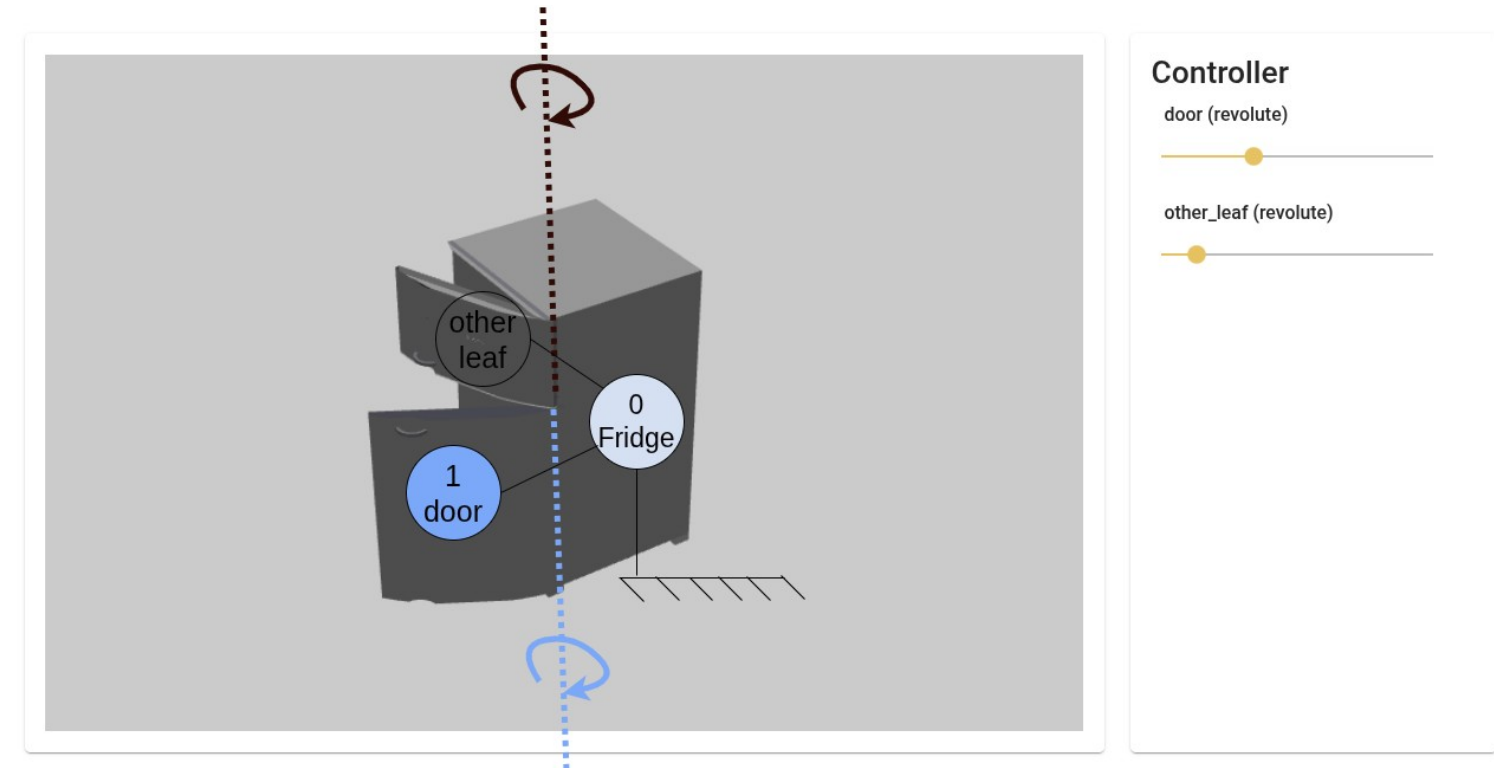


DoF : Degree of Freedom

Object-centric environment representation



Object-centric
representation



Articulated Object open source dataset of Articulated object
from UC San Diego University <https://sapien.ucsd.edu/browse>

- Environment modelisation : Open Kinematic Chain
- Domestic environments : contains articulated object with at least one degree of freedom. (mechanism with several rigid parts linked by joints)

[1] F. Xiang et al., "SAPIEN: A Simulated Part-based Interactive Environment," Mar. 19, 2020, arXiv: arXiv:2003.08515. doi: 10.48550/arXiv.2003.08515.

Problematic



How can a robot learn to manipulate
any real environment?



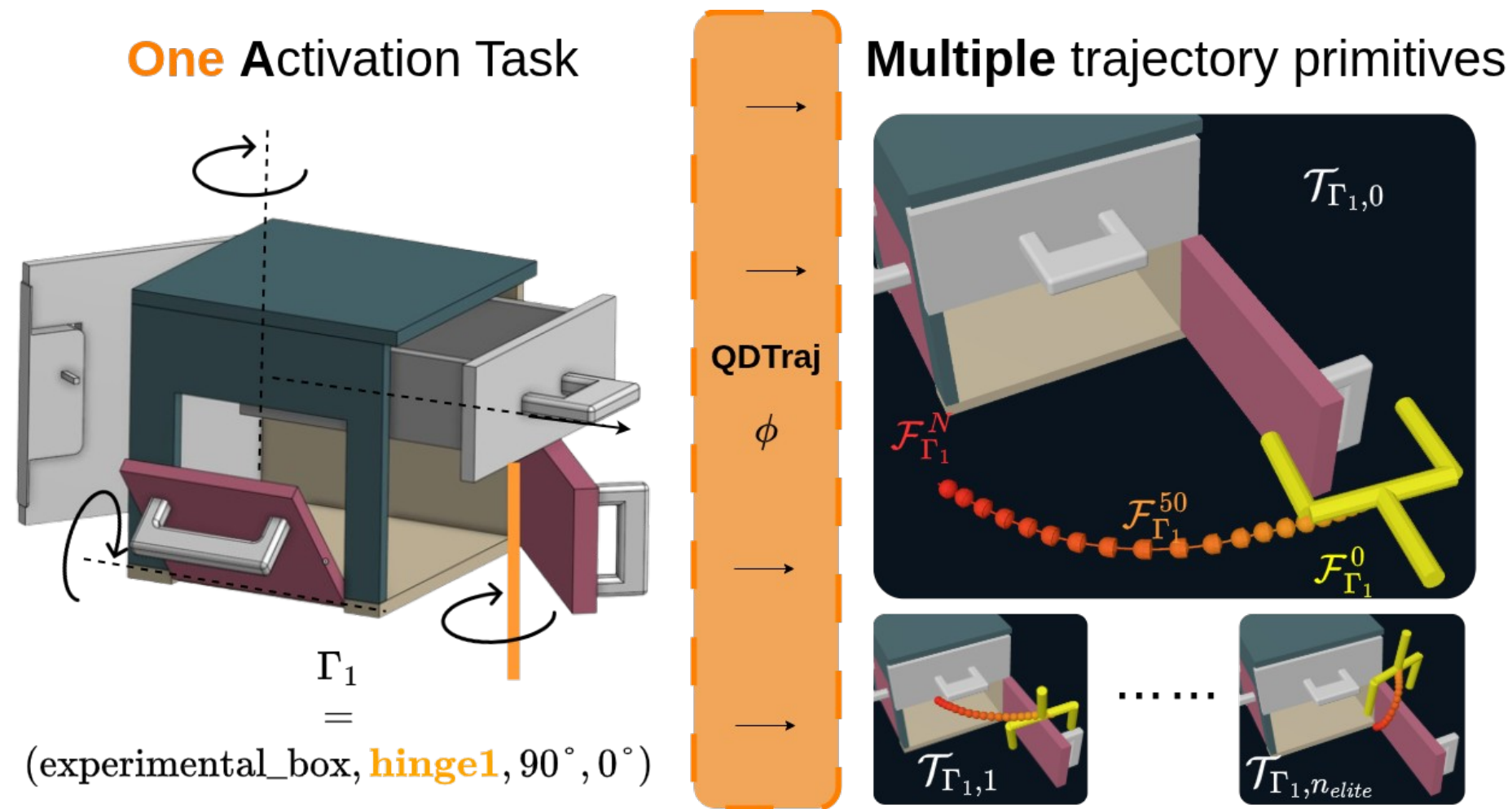
**How to manipulate each degree of freedom of the
robot real-world environment ?**

Learning expert object-centric trajectory primitives

- Input = Task primitive to manipulate a specific joint on a given object
- Output = Set of diverse trajectory primitives to manipulate a same joint.



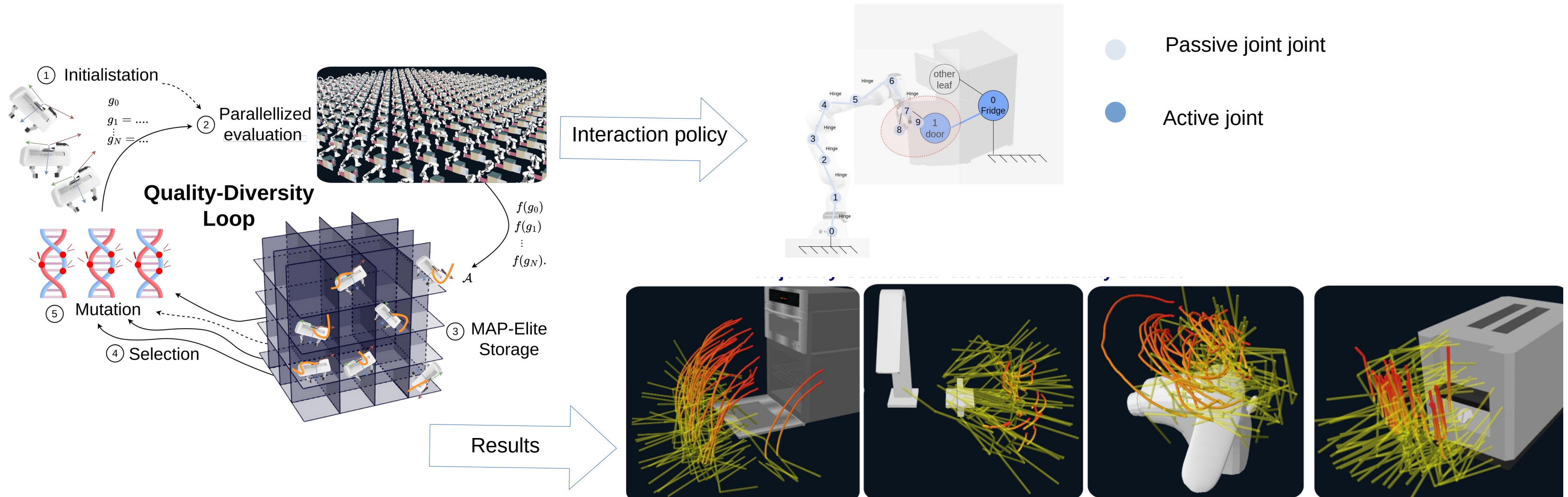
<https://genesis-world.readthedocs.io/en/latest/>



[1] M. Kappel, M. Khoramshahi, L. Annabi, F. B. Amar, and S. Doncieux, "QDTraj: Exploration of Diverse Trajectory Primitives for Articulated Objects Robotic Manipulation," Apr. 24, 2026, arXiv: arXiv:2604.22551. doi: 10.48550/arXiv.2604.22551. (IROS 2026)

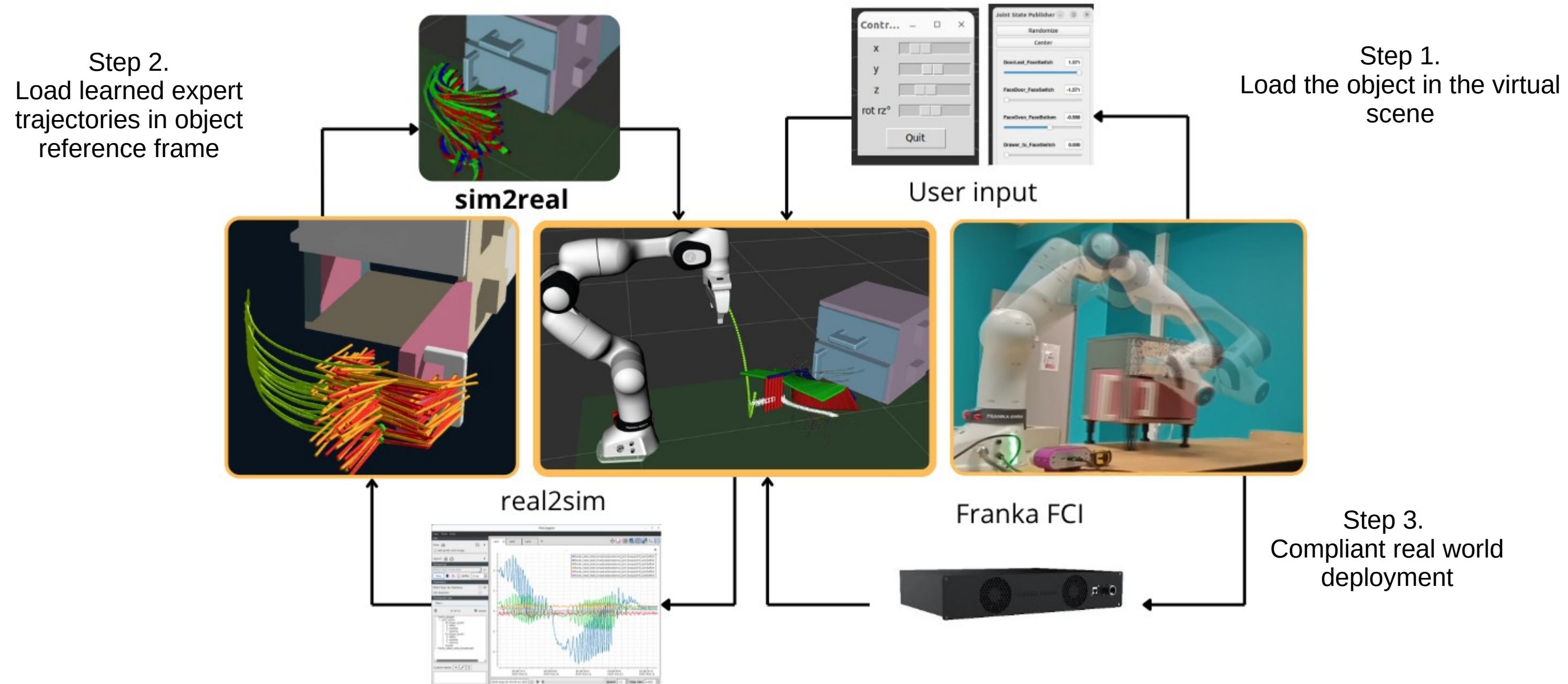
Details on the Quality-Diversity learning strategy

- Evolutionary algorithm → Exploration strategy → Exploration properties [1]
- Interaction policy = **Adaptive compliant policy**



[1] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," Apr. 20, 2015, arXiv: arXiv:1504.04909. doi: 10.48550/arXiv.1504.04909.

ROS2 Architecture for Real World Deployment

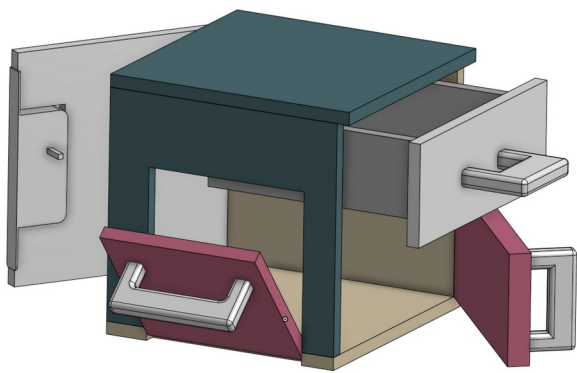


Moveit2 virtual planning scene merges Real and Simulated data

Step 1: Load the object in the virtual scene

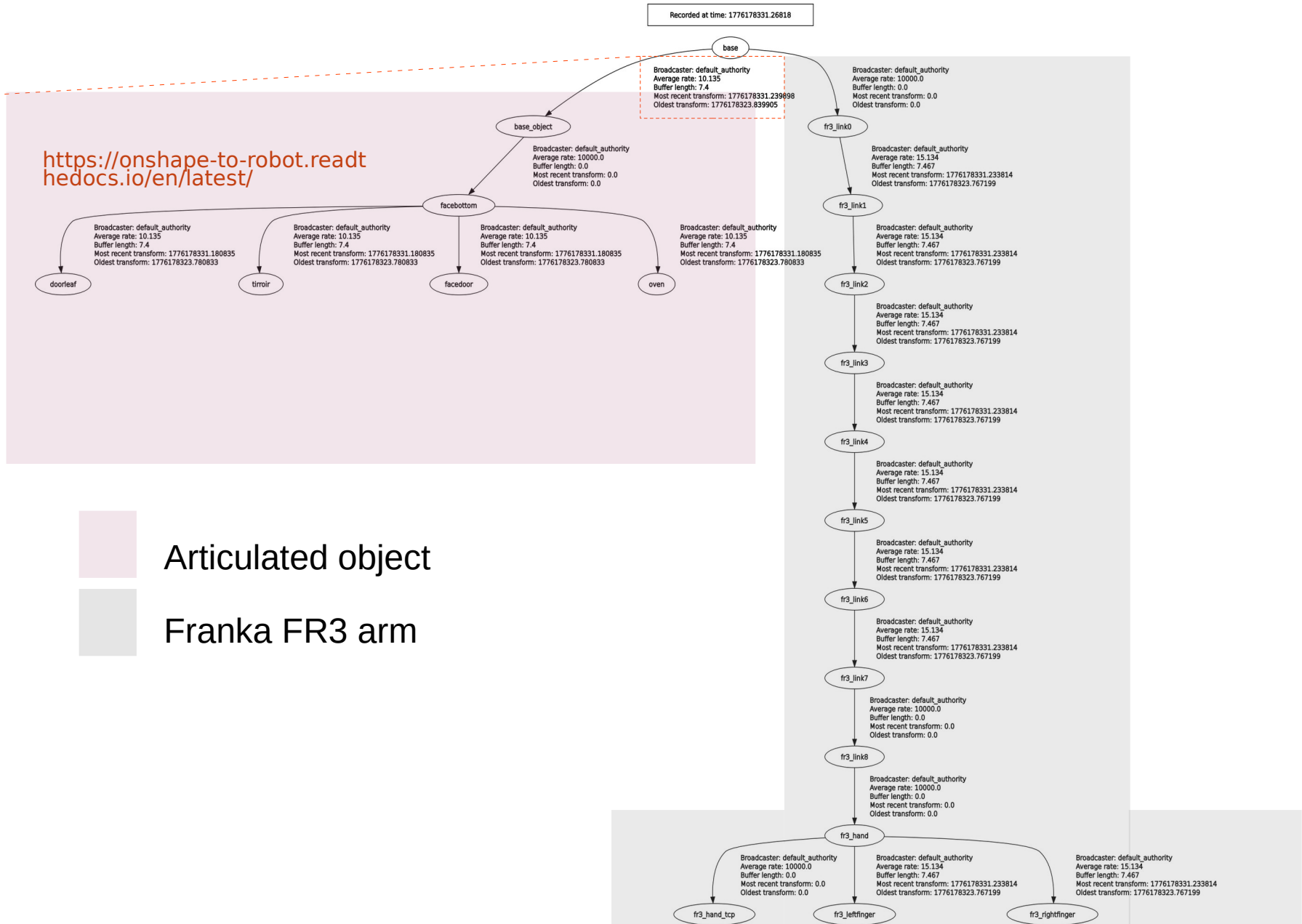
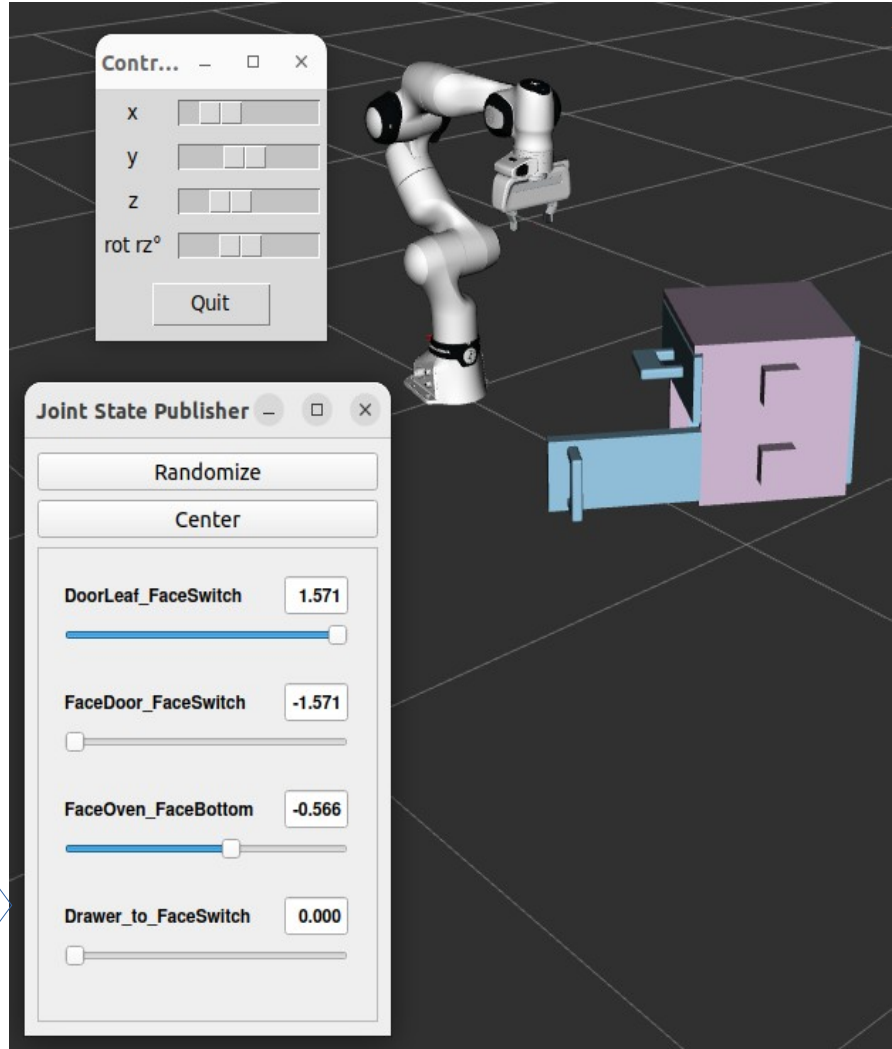


CAD
Modélisation



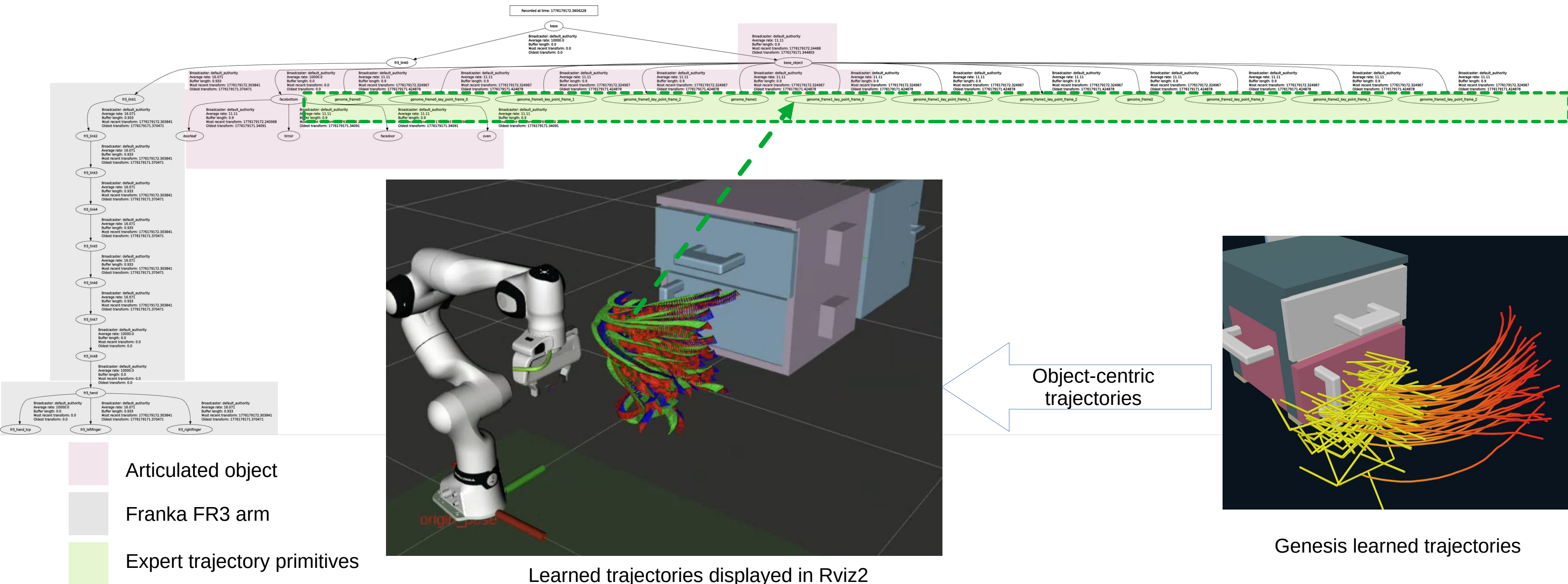
onshape

From CAD
to URDF



<https://onshape-to-robot.readthedocs.io/en/latest/>

- Node to load expert trajectories in the object reference frame



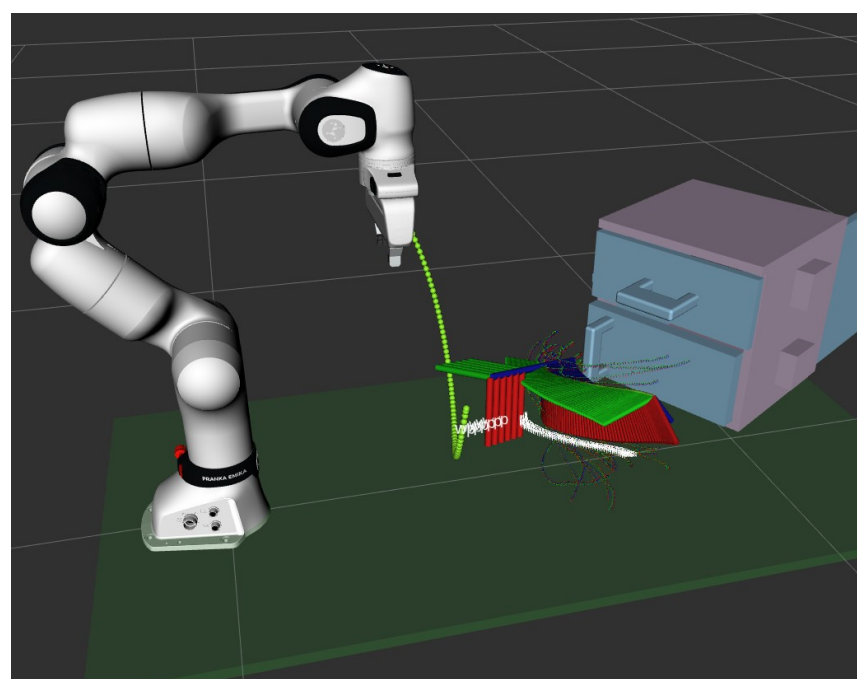
Trajectory selection among the diversity of expert primitives

If the sequence of sub-trajectories is successfully calculated in the virtual scene, the trajectory is selected for the actual deployment; otherwise, another trajectory is tested.

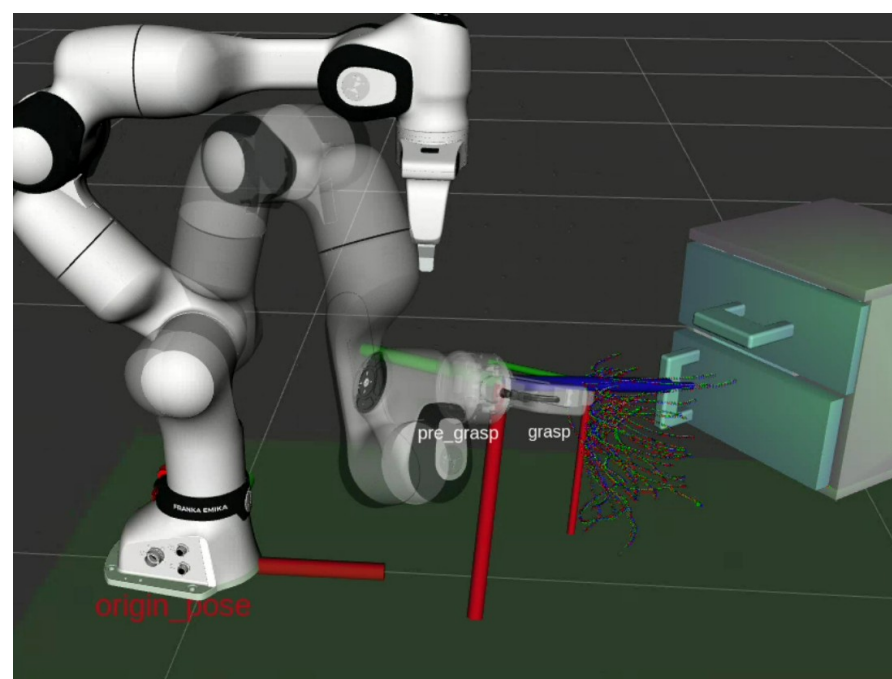
A. plan from init to pre_grasp : **Compute cartesian path** from one 6dof eef target pose

B. plan from pre_grasp to grasp : **Compute cartesian path** from straight line waypoints

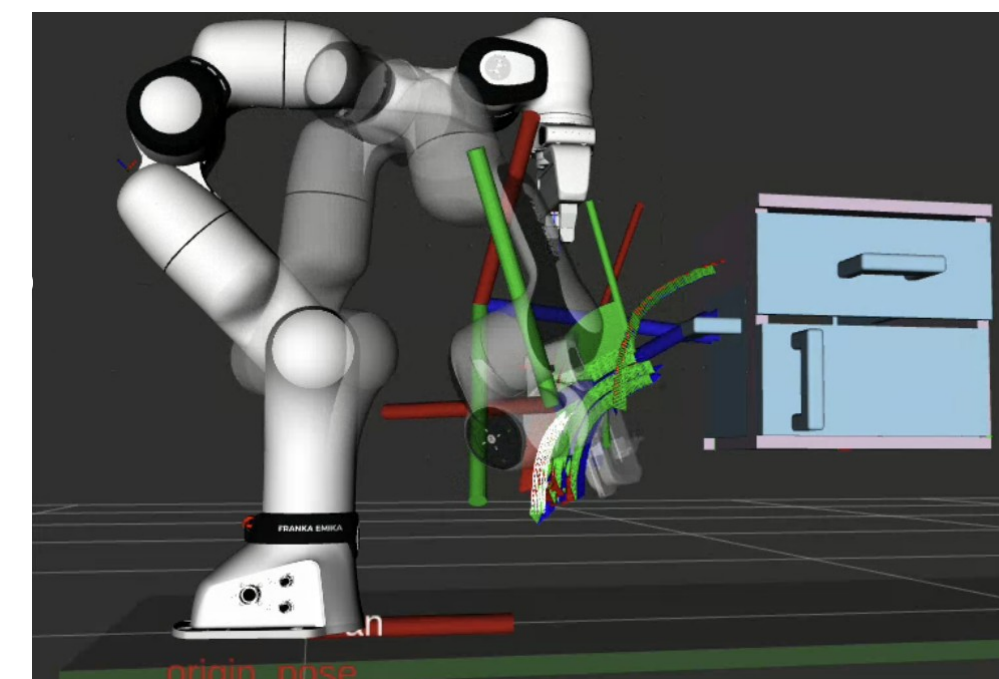
C. plan from grasp to trajectory : **Compute cartesian path** from experts waypoints of the primitive trajectories



Plan A: int → pre-grasp



Plan B : pre-grasp → grasp



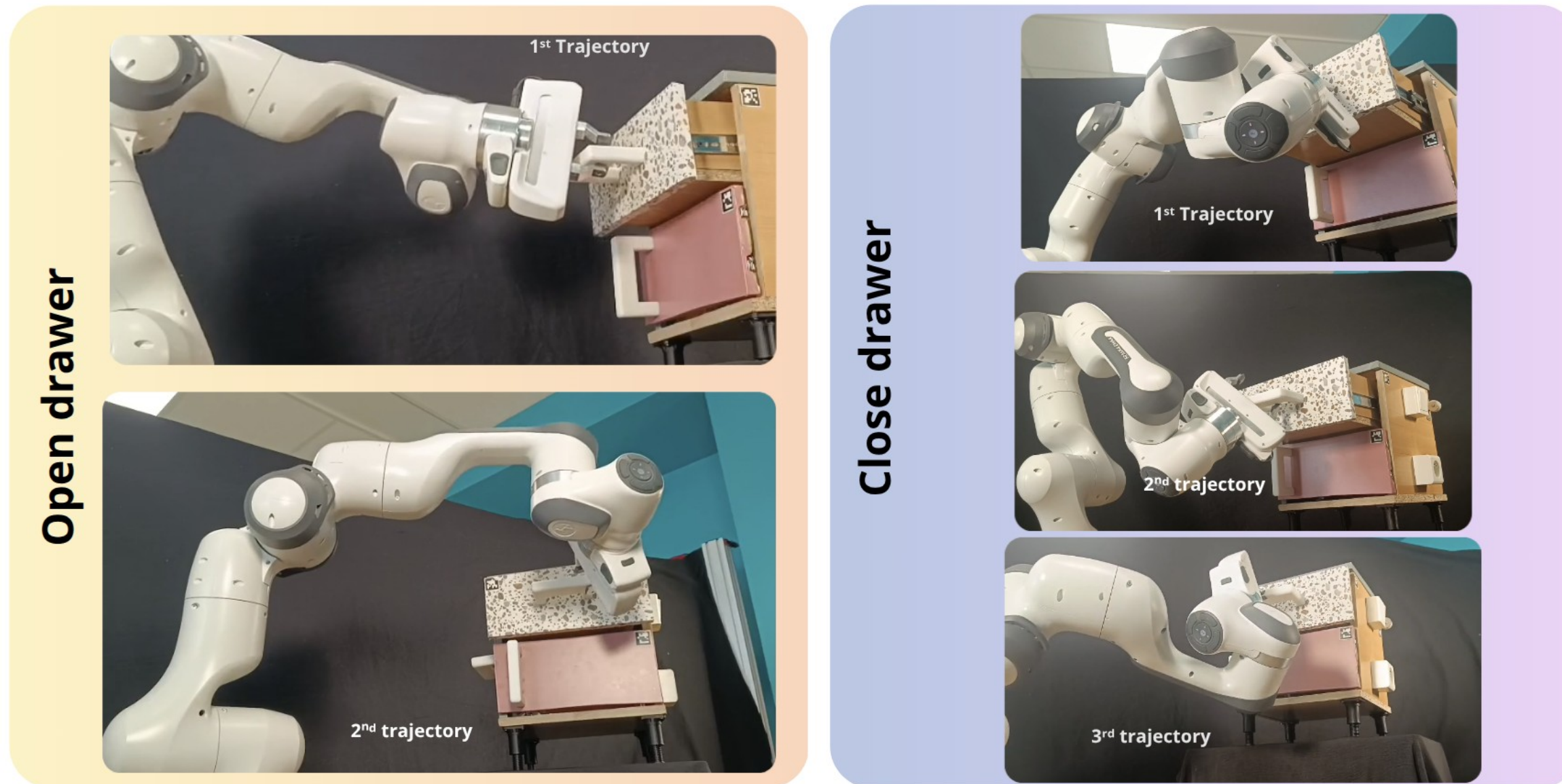
Plan C : grasp → trajectory primitive

Step 3: Compliant Real World Deployment

- FR3 uses ROS2 FollowJointTrajectory controllers to execute **cartesian path** computed in the virtual scene planning.

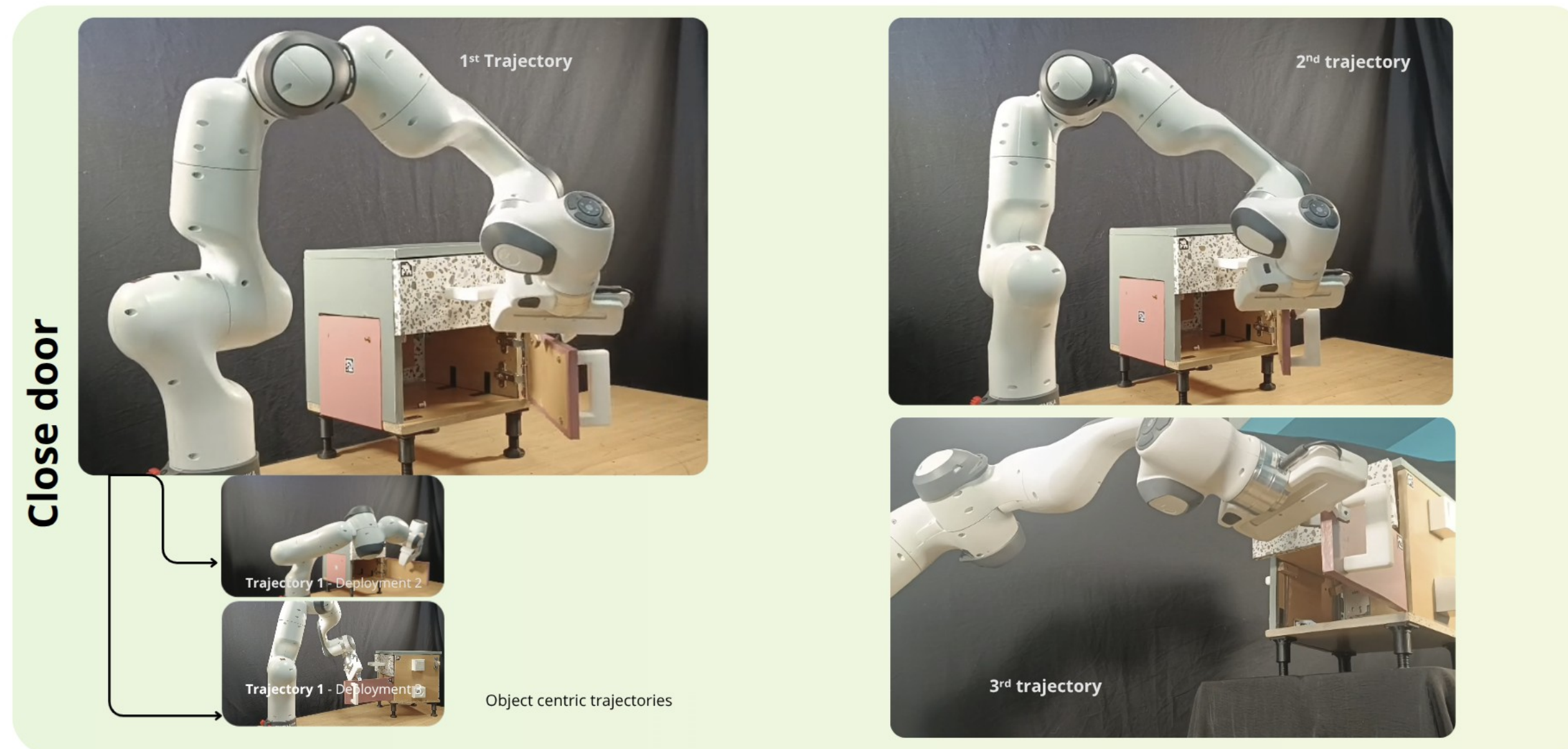
Slider activation task

MAP-Elite-Explore-Adaptive outcome deployment



Step 3: Compliant Real World Deployment

Hinge activation task MAP-Elite-Explore-Adaptive outcome deployment



Conclusion and observations

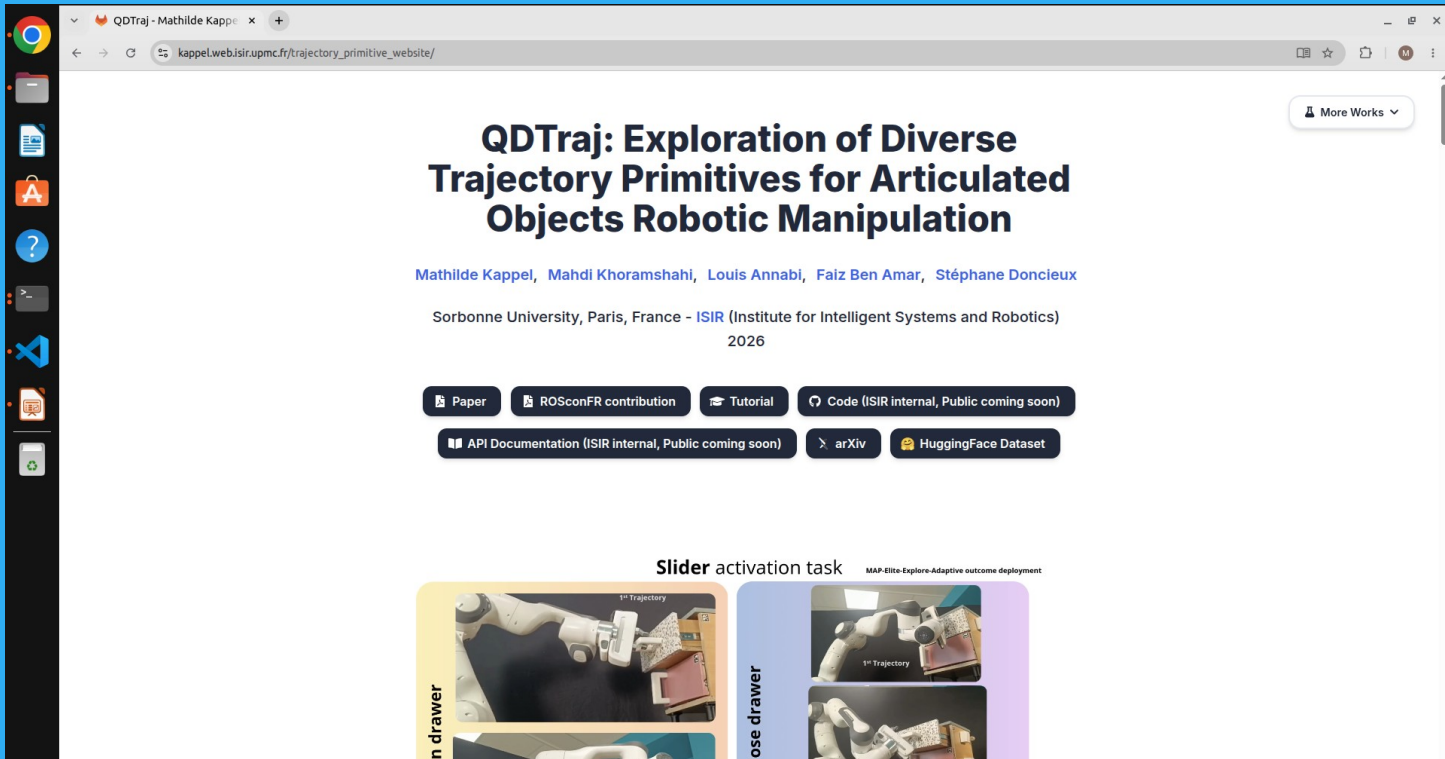


How can a robot learn to manipulate any real environment?

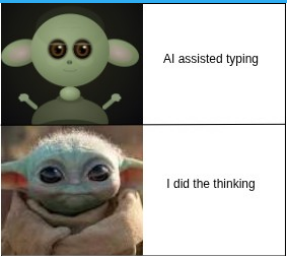
- **Autonomous manipulation of an experimental articulated object** was carried out on an FR3 robot in a real-world scenario.
- The **diversity** of the trajectory primitives learned during simulation enabled the robot to adapt to the various real-world configurations tested, thereby reducing planning failures.
- MoveIt's planning **tolerance** proved to be a key factor in the trajectory selection stage.
- The use of the **compliant controller** facilitated trajectory tracking by absorbing disturbances caused by real-world interactions with the articulated object
- Limits and future works : Incorporating torque or force feedback into the end-effector would allow for better management of contact interfaces

Project webpage:
https://kappel.web.isir.upmc.fr/trajectory_primitive_website/

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